Humanities (History & Geography)

Year 8

Cambridge Lower Secondary

Student Manual

Prime Books · Humanities (History & Geography) Year 8 · Student Book

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Unit 1 also includes a Final project, Check your progress, Evaluation, What can you do, and a Key words round-up.

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INTRODUCTION

How this book works

Geography is the study of the world and our place in it, and geographers do not just read about places: they go there. In this first unit you will learn how real fieldwork is planned and carried out, how to ask a question that data can actually answer, and how GIS turns your findings into a map that reveals patterns.

What makes this book different

- **Everything is original.** Every sentence, example and diagram was made for this book.
- **Every claim is checked.** Facts come from real, cited sources; every numerical answer was executed by computer before publication.
- **Thinking is taught explicitly.** Dedicated panels show you how specialists in this subject actually reason.

A note to teachers and parents

This book follows the spirit of an international lower-secondary programme, interpreted afresh for Prime School. Each topic opens with clear outcomes and closes with self-assessment, so progress is visible to pupil and teacher alike.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
Learning outcomes	What you will be able to do by the end.
Getting started	A warm-up that wakes up what you know.
Key words	The vocabulary of the topic, defined.
Source	A photograph, map or document to study.
Your turn	Questions and activities for you to do.
Think like a geographer	A task that builds geographical thinking.
Summary checklist	'I can' statements to check your progress.

How to work

Read with a pencil in your hand. When the book asks a question, try to answer it before reading on. Do the work in your notebook, and check your reasoning, not just your answer.

GETTING SET UP

Before Unit 1

You need a pencil, a ruler, a notebook and your curiosity. Anything else this unit requires is listed where it is needed.

Your toolkit

- A notebook for working, planning and recording, not just for final answers.
- The habit of checking: recalculate, reread, and ask whether the answer is sensible.
- The confidence to say "I am not sure yet", which is where all real learning starts.

A quick self-check

1. What do you already know about this topic?
2. Which part of it do you expect to find hardest, and why?
3. How will you know when you have genuinely understood something, rather than just recognised it?

Unit 1 begins now. Work carefully, check honestly, and ask good questions.

UNIT 1

1

Fieldwork and GIS

Going out to find the answer, and mapping what you find

IN THIS UNIT, YOU WILL

- ✓ Explain what fieldwork is and why geographers do it.
- ✓ Turn a curiosity into a good fieldwork question.
- ✓ Collect data honestly, and know its limitations.
- ✓ Present findings using maps and GIS.

✦ GETTING STARTED

Geographers do not just read about places. They go there. In pairs, discuss:

- what could you learn about your street by standing in it that you could never

learn from a book?

- what could a book tell you that standing there could not?
- what would you need to write down, so someone else could trust your findings?

UNIT 1 · TOPIC 1.1

What is fieldwork?

Learn

Fieldwork is geography done outside the classroom: going to a real place to collect your own **primary data**. Data someone else collected is **secondary data**, and both are useful, but they answer different questions.

A geographer might count vehicles at a junction, measure the width of a stream, survey shoppers about where they travelled from, or record how land is used along a street. Every one of these produces evidence you gathered yourself and can defend.

❝ SOURCE A · A REAL FIELDWORK SETTING

Scientists working on Portugal's coastline carry out fieldwork under water, surveying rocky reefs and seagrass meadows off the Algarve. They swim fixed transect lines, recording every species they see on a waterproof slate, at the same depth and season each year, so that one year can honestly be compared with the next.

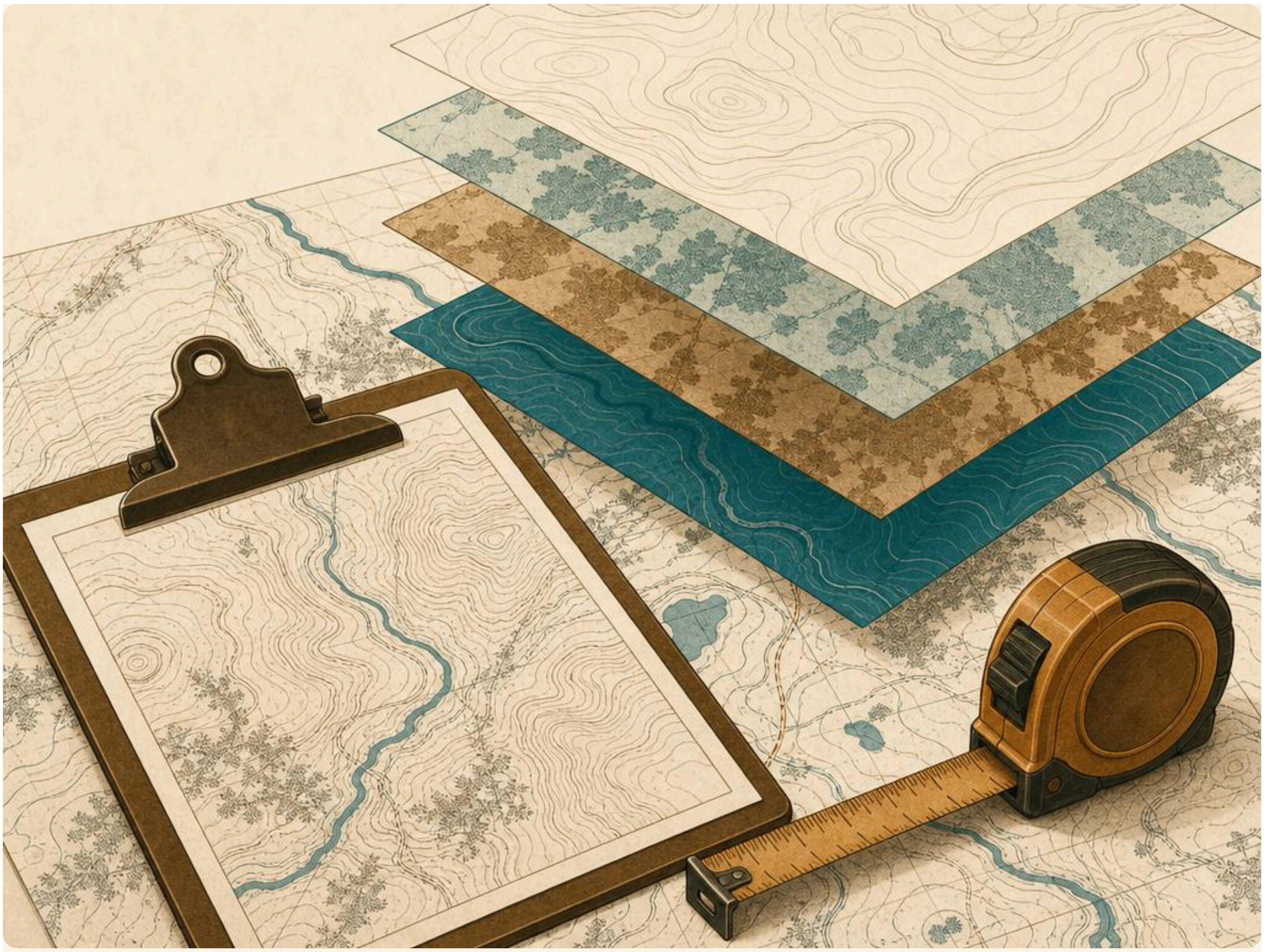

Figure 1.1 Fieldwork under water: recording data on a fixed transect.

YOUR TURN

1. What is the difference between primary and secondary data? Give an example of each.
2. Look at Source A. Why do the surveyors return at the same depth and in the same season each year?
3. Suggest two things that could go wrong with underwater data collection.
4. Give one geographical question that could **only** be answered by going outside.

UNIT 1 · TOPIC 1.2

Asking a fieldwork question

Learn

A fieldwork question must be **answerable with data you can actually collect**. "Is my town nice?" cannot be answered. "Does traffic on Rua da Igreja get busier between 08:00 and 09:00?" can.

A good fieldwork question is:

- **specific** about place, time and what is measured
- **measurable**, so two people would record the same thing
- **answerable** in the time and space you have
- **worth asking**, so the answer tells you something real

YOUR TURN

1. Explain why "Is my town nice?" is a poor fieldwork question.
2. Improve each of these into a good fieldwork question:
   a. Are there lots of cars?
   b. Do people like the park?
   c. Is the beach dirty?
3. Write a fieldwork question about your own school grounds, and say exactly what you would measure to answer it.

UNIT 1 · TOPIC 1.3

Collecting data

Learn

Data is only as good as the method behind it. Geographers plan **how**, **where** and **when** before going out, and record the plan so others can check it.

METHOD	WHAT IT COLLECTS	A LIMITATION
tally count	Numbers of things passing a point	Only one point, only one time
questionnaire	People's views and behaviour	People may not answer honestly
field sketch	The look and layout of a place	Depends on the observer
measurement	Widths, depths, gradients	Equipment can be inaccurate
land-use survey	How each building is used	Uses may be mixed or unclear

Every method has limitations, and a good geographer states them. That is not weakness. Naming your limitations is what makes your conclusion trustworthy.

● YOUR TURN

1. Choose the best method for each enquiry, and give a reason:
   a. How many pupils cycle to school?
   b. Why do people choose one supermarket over another?
   c. How does the width of a stream change downstream?
2. A student counts cars for five minutes at 11:00 on a Sunday and concludes the road is quiet. Give two reasons why this conclusion is unsafe.
3. Design a tally sheet for counting how people arrive at your school. Show the categories you would use.

∴ THINK LIKE A GEOGRAPHER

Two groups survey the same street and get different results. List three possible reasons, and explain how the groups could work out which result to trust.

UNIT 1 · TOPIC 1.4

Mapping with GIS

Learn

GIS stands for Geographic Information System: software that stores data against a location, so it can be mapped, layered and analysed. Your phone's map app is a GIS. So are the systems that route ambulances and plan flood defences.

The power of GIS is **layers**. Put your traffic counts on one layer, schools on a second and accident records on a third, and patterns appear that no single layer would reveal.

GIS CAN	EXAMPLE
map your own data	Plot your tally counts at each junction
layer datasets	Compare flood risk with where houses are
measure	Find the true distance along a winding road
analyse	Identify which areas are furthest from a clinic

● YOUR TURN

1. What does GIS stand for, and what does it do?
2. Explain why layers make GIS more powerful than a single printed map.
3. Give one real decision a town council could make better using GIS.
4. You have surveyed litter at ten points in a park. Describe how you would present this using GIS, and what pattern you would look for.

☑ SUMMARY CHECKLIST

- ✓ I can explain what fieldwork is and distinguish primary from secondary data.
- ✓ I can write a specific, measurable fieldwork question.
- ✓ I can choose an appropriate method and state its limitations.
- ✓ I can explain how GIS uses layers to reveal patterns.

UNIT 1 · FINAL PROJECT

A real fieldwork enquiry

★ FINAL PROJECT

A real fieldwork enquiry

Plan and carry out a genuine fieldwork enquiry in your school grounds or immediate neighbourhood, then present your findings as a mapped report.

What to hand in

- Your fieldwork question, specific and measurable.
- Your method, including how, where and when you collected data.
- Your raw data, presented in a table.
- A map of your findings, and a paragraph on the limitations of your data.

How it will be judged

| Criterion | Weight | |---|---| | Quality of the fieldwork question | 25% | | Rigour of the method | 25% | | Accuracy and honesty of the data | 25% | | Mapping and limitations | 25% |

UNIT 1 · CHECK YOUR PROGRESS

1. Explain the difference between primary and secondary data, with an example of each. [2]
2. Why is 'Is my town nice?' a poor fieldwork question? Rewrite it well. [2]
3. Name three data-collection methods and one limitation of each. [3]
4. A student counts cars for five minutes at 11:00 on a Sunday and concludes the road is quiet. Give two reasons this is unsafe. [2]
5. What does GIS stand for, and why are layers so powerful? [2]
6. Give one real decision a council could make better using GIS. [1]

UNIT 1 · EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
explain what fieldwork is and distinguish primary from secondary data	○	○	○
write a specific, measurable fieldwork question	○	○	○
choose a method and state its limitations	○	○	○
explain how GIS uses layers to reveal patterns	○	○	○

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- ☐ Explain what fieldwork is and why geographers do it.
- ☐ Write a specific, measurable fieldwork question.
- ☐ Choose appropriate methods and state their limitations.
- ☐ Explain how GIS layers reveal patterns.

GLOSSARY

Unit 1

fieldwork: geography carried out in a real place, outside the classroom.

primary data: data you collected yourself.

secondary data: data collected by someone else.

questionnaire: a set of questions used to collect people's views.

GIS: Geographic Information System: software that maps and analyses data by location.

transect: a fixed line along which data is collected.

SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

§ UNIT 1 · FIELDWORK AND GIS

1. **Underwater fieldwork on the Portuguese coast. Marine surveys record species along fixed transects at consistent depth and season.**
   https://en.wikipedia.org/wiki/Transect

2. **Geographic Information Systems. How GIS stores, layers and analyses spatial data.**
   https://en.wikipedia.org/wiki/Geographic_information_system

PRIME BOOKS · YEAR 8

Go there. Measure it. Map it.

This Year 8 Student Book puts you in the field. You will learn what makes a fieldwork question answerable, choose methods and state their limitations honestly, and use GIS layers to reveal patterns no single map could show.

Written for Prime School pupils, in clear British English, with real places and real methods.

FIELDWORK | DATA | METHODS | GIS | MAPPING

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